

The Generalized Focal Loss: A Saturating Two-Parameter Family with Bounded Influence for Network Intrusion Detection

Shahmeer Ali, Ali Rauf, Mohammad Soban
 Department of Artificial Intelligence
 FAST-NUCES Islamabad
 {i230119, i230076, i230056}@isb.nu.edu.pk

Abstract—We analyze the focal loss of Lin *et al.* as an M-estimator and identify two structural defects: an unbounded influence function as the predicted confidence $p_t \rightarrow 0^+$, and a homogeneous-modulator coupling that ties easy-example suppression to hard-example amplification through a single scalar γ . We propose a two-parameter generalization, the *Generalized Focal Loss* (GFL), defined by replacing the polynomial modulator $(1 - p_t)^\gamma$ with the Hill-type saturating kernel $\phi_{\gamma,\tau}(p_t) = (1 - p_t)^\gamma / [1 + \tau(1 - p_t)^\gamma]$. We prove that (i) GFL recovers focal loss exactly at $\tau \rightarrow 0$ and cross-entropy at $\gamma \rightarrow 0$; (ii) the modulator is uniformly bounded above by $1/(1 + \tau)$ on $[0, 1]$; (iii) the per-sample logit gradient at the noise boundary is attenuated relative to focal loss by exactly $1/(1 + \tau)$; (iv) a q -deformed extension achieves uniformly bounded influence on the closed interval $[0, 1]$. We validate the bounded-influence theorem by direct gradient-norm measurement: on synthetic logits the predicted attenuation factor matches observation to $3 \times 10^{-4}\%$ relative error across $\tau \in [0, 5]$; on real CICIDS2017 samples produced by a CE-warmup baseline the prediction matches observation to within 10% relative error. Downstream F1 on CICIDS2017 with the bounded-influence loss family is dataset-limited: the moderate-rare class tier (training counts 400–5000) shows the predicted noise-as-regularization signature for the q -deformed variant, with mean F1 increasing by $+0.106$ from clean to 20% BENIGN-to-minority label noise, while standard CE, FL, and GFL show degradation slopes within ± 0.013 on the same tier. We provide the construction, proofs, validation methodology, and CICIDS2017 evaluation, with honest reporting of where the F1 metric does and does not reflect the predicted gradient-level effect.

Index Terms—focal loss, label noise, bounded influence, intrusion detection, CICIDS2017, imbalanced classification, robust learning

I. INTRODUCTION

Deep-learning network intrusion detection systems on CICIDS2017 [3] routinely report aggregate accuracy above 99%, yet the operationally important attack classes remain hard to detect. Heartbleed appears 11 times in 2.8M flows; Infiltration appears 36 times. A model that flags every benign flow correctly while missing the 11 Heartbleed records achieves the same headline accuracy as one that catches all 11—and the difference between those two systems is the entire reason the task exists.

The dominant technique for addressing this imbalance has been data-level rebalancing: SMOTE [2] synthesizes minority samples before training. SMOTE is effective in moderate-

imbalance regimes [18] but provides no signal on the extreme tail where there are essentially no samples to interpolate between. The loss-level alternative is focal loss [1], originally proposed for object detection, which down-weights well-classified examples through the polynomial modulator $(1 - p_t)^\gamma$. Adoption in intrusion detection has been steady, but its reliability has been mixed; recent surveys report focal loss helping on some attack types and degrading on others without a clear pattern [13], [14].

This paper begins by treating focal loss as an M-estimator and asking what its functional form actually does. Two structural defects emerge from the asymptotic analysis (Section IV). First, the loss is *unbounded in influence*: as $p_t \rightarrow 0^+$ the per-sample gradient diverges as $\Theta(1/p_t)$, so a single mislabeled rare-class flow can dominate a minibatch’s gradient. Second, the modulator is *globally homogeneous*: the single scalar γ controls both how aggressively easy examples are suppressed and how strongly hard examples are amplified, and these cannot be separated.

We propose a two-parameter generalization that fixes both defects. The *Generalized Focal Loss* (GFL, Section V) replaces the polynomial modulator with the Hill-type saturating kernel

$$\phi_{\gamma,\tau}(p_t) = \frac{(1 - p_t)^\gamma}{1 + \tau(1 - p_t)^\gamma},$$

parametrized by a focusing exponent γ and an independent saturation scale τ . The construction recovers focal loss exactly at $\tau \rightarrow 0$ and cross-entropy at $\gamma \rightarrow 0$, so nothing is lost relative to existing baselines. What is gained is provable: the modulator is uniformly bounded above by $1/(1 + \tau)$ on $[0, 1]$, the divergent gradient at $p_t \rightarrow 0^+$ is attenuated by exact factor $1/(1 + \tau)$ (Section VII), and a q -deformed extension achieves uniformly bounded influence on the closed interval (Section X).

We validate the central bounded-influence theorem directly, by measuring per-sample gradient norms (Section XII). On synthetic logits across the full range $p_t \in [10^{-9}, 0.999]$, the predicted gradient-attenuation factor $1/(1 + \tau)$ matches observation to $3 \times 10^{-4}\%$ relative error across all tested $\tau \in \{0, 0.3, 0.7, 1, 1.5, 2, 5\}$. On real CICIDS2017 samples produced by a CE-warmup MLP, the prediction holds to within 10% relative error—the loosening attributable to finite-sample

averaging over the 80 real samples that fall in the relevant low- p_t regime.

We complement the gradient-level validation with downstream F1 evaluation on CICIDS2017 (Section XIII). The headline finding is dataset-limited: only 0.4% of test-time samples produced by a converged classifier fall in the regime where the bounded-influence advantage substantially modifies the gradient. Consequently, F1 differences between FL and standard GFL are within seed variance. The q -deformed extension does produce a measurable F1 effect on a moderate-rare class tier (training counts 400–5000), where it exhibits the noise-as-regularization signature predicted by Theorem 16: F1 *increases* from 0.637 at clean labels to 0.743 at 20% BENIGN-to-minority noise, while CE, FL, and GFL show small degradations on the same tier.

a) Contributions.:

- A two-parameter loss family (Definition 6) with closed-form derivatives in both probability and logit, recovering FL and CE as parametric limits.
- Asymptotic analysis (Lemma 1–Lemma 4) formalizing two structural defects of standard FL.
- A quantitative gradient-attenuation theorem (Theorem 11) and a uniformly-bounded q -deformed extension (Theorem 16).
- Direct numerical validation of the central theorem on synthetic logits ($3 \times 10^{-4}\%$ error) and real CICIDS2017 samples ($\leq 10\%$ error).
- Honest downstream evaluation on CICIDS2017 with tier-stratified per-class F1, identifying where the predicted effect manifests in F1 and where it does not.

II. BASE PAPER AND DOMAIN-TRANSFER REPLICATION

A. Base Paper: Focal Loss (Lin *et al.*, ICCV 2017)

We formally adopt Lin *et al.* [1] as the base paper for this work. Lin *et al.* introduced focal loss to address the extreme foreground–background class imbalance inherent in one-stage dense object detection, where the training set contains vastly more easy background examples than foreground objects. Their core contribution is the polynomial modulator $(1 - p_t)^\gamma$ applied to cross-entropy, which reduces the loss contribution of well-classified examples so that training focuses on the hard-to-classify tail. On the COCO detection benchmark using the RetinaNet architecture with a ResNet-101-FPN backbone, they report a single-model average precision of 39.1%, outperforming all prior two-stage detectors at comparable inference speed. The recommended default hyperparameters are $\gamma = 2$ and a class-prior-initialized output bias ($\pi = 0.01$).

B. Summary of Base Paper Methodology

The focal loss modifies cross-entropy as:

$$\mathcal{L}_{\text{FL}}(p_t) = -\alpha_t(1 - p_t)^\gamma \log p_t, \quad \gamma \geq 0,$$

where p_t is the predicted probability for the ground-truth class and α_t is a class-weighting scalar. At $\gamma = 0$ this reduces to weighted cross-entropy; at $\gamma = 2$, a well-classified example with $p_t = 0.9$ contributes a loss $100\times$ smaller than under

TABLE I
PILOT ARCHITECTURE ABLATION ON CICIDS2017. ALL VARIANTS USE $\gamma = 2$ WHERE FOCAL LOSS IS APPLIED. BEST MACRO-F1 IN BOLD.

Variant	Macro-F1	False-alarm rate
Full (CNN+Attn+LSTM+FL)	0.7938	0.0199
w/o Attention	0.8010	0.0200
w/o LSTM (CNN+Dense+FL)	0.8137	0.0148
w/o CNN (LSTM+FL)	0.5407	0.0180
Cross-entropy ($\gamma=0$)	0.8094	0.0155

cross-entropy. The modulator concentrates gradient mass on the hard-negative tail of the training distribution—the core mechanism we replicate and subsequently analyze.

C. Domain-Transfer Replication on CICIDS2017

Reproducing Lin *et al.* on COCO requires training large-scale convolutional object detectors on a 330k-image benchmark, which is computationally prohibitive and architecturally orthogonal to network intrusion detection. The accepted academic practice for cross-domain adoption is *domain-transfer replication*: apply the base paper’s method to the target domain and verify that the core claimed mechanism holds. We apply focal loss at the recommended $\gamma = 2$ to CICIDS2017 and measure the focusing effect directly.

Our main evaluation (Section XIII, Table IV) confirms the mechanism: FL achieves 0.775 macro-F1 versus CE at 0.706 (+0.069 gain), and minority-F1 rises from 0.565 to 0.670. These results are consistent with Lin *et al.*’s claim that focal loss concentrates learning on hard, minority-class examples. This domain-transfer replication (i) establishes focal loss as a valid and effective baseline in the NIDS domain, and (ii) frames the theoretical critique of Section IV: *even though the focusing mechanism works empirically, its two structural defects motivate the GFL generalization.*

III. PILOT STUDY AND ARCHITECTURAL MOTIVATION

Prior to formulating the loss-function contribution, we conducted an architecture-ablation pilot using the dominant design pattern in deep-learning IDS literature: a CNN front-end followed by sequence and attention components, trained with focal loss. Table I reports macro-F1 and benign-traffic false-alarm rate across architectural variants on a CICIDS2017 subset. The results establish three findings that, together, redirect the project from architectural innovation to loss-function design.

The ablation produced three findings that shaped the rest of this work.

a) *Finding 1: CNN is the sole load-bearing component.*: Removing the CNN front-end (LSTM+FL, row 4) collapses macro-F1 from 0.8137 to 0.5407—a 0.27 drop, an order of magnitude larger than any other architectural change in the table. The recurrent and attentional modules cannot recover the discriminative information that the CNN was extracting. This identifies a basic feature extractor as a necessary component, while every other element is candidate for removal.

b) *Finding 2: Architectural complexity beyond CNN degrades performance.*: The full pipeline (CNN+Attn+LSTM+FL) scores 0.7938. Removing attention raises it to 0.8010. Additionally removing the LSTM raises it further to 0.8137. Each component added on top of the CNN backbone made the model worse, not better, despite each being individually well-motivated by prior IDS literature. The monotone improvement under stripping—rather than under adding—is the empirical signature of an over-parameterised architecture for the available data scale, and it directly contradicts the assumption that more sophisticated sequence and attention machinery yields better minority-class recall.

c) *Finding 3: Focal loss did not improve over cross-entropy.*: The best focal-loss variant (CNN+Dense+FL, row 3) achieves 0.8137 macro-F1 versus 0.8094 for cross-entropy on the same architecture (row 5)—a gap of 0.0043, well within the seed-to-seed standard deviation we measure in §XI (≤ 0.0032 for any single configuration). The focal loss did not provide a meaningful improvement on this task. False-alarm rates tell the same story: 0.0148 for FL versus 0.0155 for CE, statistically indistinguishable. Whatever mechanism focal loss is supposed to provide for imbalanced classification is not surfacing on CICIDS2017 with this architecture.

d) *Methodological consequences.*: We draw two methodological consequences from these findings. First, the empirical bottleneck on this dataset is not architectural—once a basic feature extractor is in place, complexity beyond it provides no measurable benefit and the loss function becomes the right axis for further work. Concretely, the 0.0043 FL-vs-CE gap is too small to motivate further architectural exploration but raises a sharper question: *why does the standard tool for imbalanced classification fail to help on a benchmark designed to expose imbalance?* This question motivates the asymptotic analysis of §IV, which traces the failure to two structural defects in the focal modulator.

Second, we adopt a multilayer perceptron (MLP) backbone for all subsequent loss-function experiments. CNN is methodologically dubious on tabular flow features, since 1D convolution presupposes meaningful adjacency between adjacent feature columns, which the bag-of-statistics representation produced by CICFlowMeter [17] does not exhibit. The MLP is the principled architecture for unordered tabular features and isolates the loss function as the only experimental variable. We note that the FL-vs-CE result *does* reverse on the MLP backbone (FL 0.775 vs CE 0.706 in Table IV)—the focusing effect of focal loss surfaces on the simpler architecture where the more complex CNN+LSTM was apparently masking it. We return to this reconciliation in §XIII-C.

IV. ASYMPTOTIC ANALYSIS OF FOCAL LOSS

Throughout, K -class probabilistic classification is performed by a parametric model $f_\theta : \mathcal{X} \rightarrow \Delta^{K-1}$, with predicted distribution $p = f_\theta(x)$ and confidence on the ground-truth class $p_t = p_y$. We write $u = 1 - p_t$ for brevity. The cross-entropy loss is $\mathcal{L}_{\text{CE}}(p_t) = -\log p_t$, equivalent to

the Bregman divergence $D_\Psi(e_y \| p)$ with negentropy generator $\Psi(q) = \sum_i q_i \log q_i$ [5]. The focal loss [1] is

$$\mathcal{L}_{\text{FL}}(p_t) = -\alpha_t(1 - p_t)^\gamma \log p_t, \quad \gamma \geq 0. \quad (1)$$

A. Three-regime asymptotics

Lemma 1 (Focal-loss asymptotics). *With $u = 1 - p_t$:*

$$\begin{aligned} \text{easy } (p_t \rightarrow 1^-) : \quad & \partial_{p_t} \mathcal{L}_{\text{FL}} \sim -\alpha_t(\gamma + 1)u^\gamma \rightarrow 0, \\ \text{boundary } (p_t \rightarrow \tfrac{1}{2}) : \quad & \partial_{p_t} \mathcal{L}_{\text{FL}} = O(1), \\ \text{hard } (p_t \rightarrow 0^+) : \quad & \partial_{p_t} \mathcal{L}_{\text{FL}} \sim -\alpha_t/p_t \rightarrow -\infty. \end{aligned}$$

Proof. Direct expansion using $\log p_t = \log(1 - u) = -u + O(u^2)$ for the easy regime, and $u^\gamma \rightarrow 1$ for the hard regime. $\square$

B. Defect 1: unbounded influence

Theorem 2 (Unbounded influence of focal loss). *For every $\gamma \geq 0$ and $\alpha_t > 0$, $\sup_{p_t \in (0,1)} |\partial_{p_t} \mathcal{L}_{\text{FL}}(p_t)| = +\infty$.*

Proof. $\partial_{p_t} \mathcal{L}_{\text{FL}} \sim -\alpha_t/p_t$ as $p_t \rightarrow 0^+$ by Lemma 1. $\square$

Remark 3. In the M-estimator framework [4], an estimator is B -robust iff $\sup_z \|\nabla_\theta \mathcal{L}\|$ is finite uniformly over a compact θ -set. Theorem 2 states that focal loss *fails* this property: a mislabelled rare-class sample with $p_{\bar{y}}(x) \approx 0$ contributes an unbounded gradient.

C. Defect 2: focusing–suppression coupling

Lemma 4 (Coupling). *For any $0 < p_h < \frac{1}{2} < p_e < 1$, the ratio*

$$R_\gamma(p_e, p_h) = \frac{\mathcal{L}_{\text{FL}}(p_e)}{\mathcal{L}_{\text{FL}}(p_h)} = \left(\frac{1 - p_e}{1 - p_h} \right)^\gamma \cdot \frac{\log p_e}{\log p_h}$$

is monotone decreasing in γ , with limit 0 as $\gamma \rightarrow \infty$.

Proof. The factor $\log p_e / \log p_h$ is γ -independent. Since $0 < (1 - p_e)/(1 - p_h) < 1$, the power $(\cdot)^\gamma$ is strictly decreasing in γ with limit 0. $\square$

The lemma formalizes the practical observation that increasing γ to amplify rare-class signal simultaneously suppresses every easy example, including the medium-hard regime $p_t \in [0.3, 0.6]$ that carries decision-boundary geometry. There is no scalar γ that controls these two regimes independently.

V. THE GENERALIZED FOCAL LOSS

A. Desiderata

We seek a modulator $\phi : (0, 1) \rightarrow \mathbb{R}_{\geq 0}$ parametrized by (γ, τ) satisfying:

- (D1) **Recovery.** A parameter limit recovers $(1 - p_t)^\gamma$.
- (D2) **Boundedness.** $\sup_{p_t \in [0,1]} \phi(p_t) < \infty$.
- (D3) **Easy fidelity.** $\phi(p_t) \rightarrow 0$ as $p_t \rightarrow 1^-$ at rate $\Theta(u^\gamma)$.
- (D4) **Smoothness.** $\phi \in C^2((0, 1))$ with closed-form derivatives.
- (D5) **Decoupling.** The boundary rate (D3) is controlled by γ alone; the upper bound (D2) is controlled by τ alone.

B. Why simple alternatives fail

Three naïve candidates are ruled out by these desiderata: $\min((1-p_t)^\gamma, c)$ fails D4 (kink at the clamp); $(1-p_t)^\gamma + \varepsilon$ fails D3 ($\phi \not\rightarrow 0$ at $p_t \rightarrow 1^-$); $1 - e^{-\gamma(1-p_t)}$ fails D1 (no parameter limit reduces to focal loss).

C. The Hill-type kernel

The simplest closed-form expression satisfying D1–D5 is the rational function

$$\phi_{\gamma,\tau}(p_t) := \frac{(1-p_t)^\gamma}{1+\tau(1-p_t)^\gamma}, \quad \gamma > 0, \tau \geq 0. \quad (2)$$

This is a Hill function of order γ with half-saturation at $\tau^{-1/\gamma}$. Writing $u = 1 - p_t$, $E = 1 + \tau u^\gamma$, the algebraic identity

$$E - \tau u^\gamma = 1, \quad \phi_{\gamma,\tau} = u^\gamma / E \quad (3)$$

enables the closed-form derivative calculations of Section VI.

Proposition 5 (Properties of $\phi_{\gamma,\tau}$). *For all $\gamma > 0$, $\tau \geq 0$: (i) $\phi_{\gamma,0}(p_t) = (1-p_t)^\gamma$ and $\phi_{0,\tau}(p_t) = 1/(1+\tau)$; (ii) $\phi_{\gamma,\tau}$ is monotone decreasing on $[0, 1]$ with $\phi(0) = 1/(1+\tau)$ and $\phi(1) = 0$; (iii) $\phi_{\gamma,\tau}(p_t) \sim u^\gamma$ as $p_t \rightarrow 1^-$; (iv) $\phi_{\gamma,\tau} \in C^\infty((0, 1))$.*

Definition 6 (Generalized Focal Loss). For $\gamma > 0$, $\tau \geq 0$, $\alpha_t \in (0, 1]$,

$$\mathcal{L}_{\text{GFL}}(p_t) := -\alpha_t \phi_{\gamma,\tau}(p_t) \log p_t. \quad (4)$$

Proposition 7 (Recovery limits). $\lim_{\tau \rightarrow 0^+} \mathcal{L}_{\text{GFL}} = \mathcal{L}_{\text{FL}}$ and $\lim_{\gamma \rightarrow 0^+} \mathcal{L}_{\text{GFL}} = \frac{1}{1+\tau} \mathcal{L}_{\text{CE}}$.

VI. DIFFERENTIAL CALCULUS OF GFL

Lemma 8 (Modulator derivatives). For $p_t \in (0, 1)$ and $\gamma \geq 1$,

$$\phi'_{\gamma,\tau}(p_t) = -\frac{\gamma u^{\gamma-1}}{E^2}, \quad (5)$$

$$\phi''_{\gamma,\tau}(p_t) = \frac{\gamma u^{\gamma-2}[(\gamma-1) - \tau(\gamma+1)u^\gamma]}{E^3}. \quad (6)$$

Proof. Quotient rule on $\phi = u^\gamma/E$ with $du/dp_t = -1$, simplified using identity (3). For ϕ'' , differentiate ϕ' and reduce the numerator using $E - \tau u^\gamma = 1$. $\square$

Theorem 9 (Gradient and Hessian of $\mathcal{L}_{\text{GFL}}$).

$$\partial_{p_t} \mathcal{L}_{\text{GFL}} = \alpha_t \left[\frac{\gamma u^{\gamma-1} \log p_t}{E^2} - \frac{u^\gamma}{p_t E} \right], \quad (7)$$

$$\begin{aligned} \partial_{p_t}^2 \mathcal{L}_{\text{GFL}} = -\alpha_t \left[\frac{\gamma u^{\gamma-2}((\gamma-1) - \tau(\gamma+1)u^\gamma) \log p_t}{E^3} \right. \\ \left. - \frac{2\gamma u^{\gamma-1}}{p_t E^2} - \frac{u^\gamma}{p_t^2 E} \right]. \end{aligned} \quad (8)$$

Theorem 10 (Logit-form gradient). For binary classification with $p_t = \sigma(z)$,

$$\partial_z \mathcal{L}_{\text{GFL}} = \alpha_t \left[\frac{\gamma p_t u^\gamma \log p_t}{E^2} - \frac{u^{\gamma+1}}{E} \right]. \quad (9)$$

Equation (9) is the implementation target. Both terms are bounded on $(0, 1)$, avoiding the $0 \cdot (-\infty)$ indeterminacy that arises near $p_t = 0$ when ϕ and $\log p_t$ are computed separately.

VII. GRADIENT FLOW ANALYSIS

Theorem 11 (Three-regime gradient comparison). Let $g_{\text{CE}}, g_{\text{FL}}, g_{\text{GFL}}$ denote $|\partial_{p_t} \mathcal{L}|$ for cross-entropy, focal loss, and GFL respectively.

- (a) As $p_t \rightarrow 1^-$: $g_{\text{CE}} \rightarrow 1$ while $g_{\text{FL}}, g_{\text{GFL}} \rightarrow 0$ at common rate $\Theta(u^\gamma)$.
- (b) At $p_t = 0.5, \gamma = 2$: $g_{\text{GFL}}/g_{\text{FL}} = 1/(1 + \tau \cdot 0.25)^2$, exceeding 0.79 for $\tau \leq 0.5$.
- (c) As $p_t \rightarrow 0^+$: $g_{\text{CE}}, g_{\text{FL}}, g_{\text{GFL}}$ all diverge, but $\lim_{p_t \rightarrow 0^+} g_{\text{GFL}}/g_{\text{FL}} = 1/(1 + \tau)$.

Proof. (a) From (7) the leading term is $u^\gamma/(p_t E) \sim u^\gamma$ since $E \rightarrow 1, p_t \rightarrow 1$. (b) Direct substitution of $u = 0.5, u^\gamma = 0.25$. (c) As $p_t \rightarrow 0^+$, $u^\gamma \rightarrow 1, E \rightarrow 1 + \tau$, the dominant term $u^\gamma/(p_t E) \rightarrow 1/((1 + \tau)p_t)$ versus $1/p_t$ for FL. $\square$

The economic content: GFL preserves easy-example attenuation, retains $\geq 79\%$ of focal gradient at $p_t = 0.5$ for $\tau \leq 0.5$, and strictly attenuates the divergent gradient at $p_t \rightarrow 0^+$ by exact constant factor $1/(1 + \tau)$.

VIII. CURVATURE ATTENUATION

The same factor $1/(1 + \tau)$ that controls gradient attenuation also governs the Hessian blow-up rate at the noise boundary. This is a second, independent quantity at which the saturation parameter acts, directly affecting the smoothness constant L seen by SGD.

Theorem 12 (Curvature attenuation at the noise boundary). For $\gamma > 1, \tau > 0$, as $p_t \rightarrow 0^+$,

$$\partial_{p_t}^2 \mathcal{L}_{\text{FL}}(p_t) \sim -\frac{\alpha_t}{p_t^2}, \quad \partial_{p_t}^2 \mathcal{L}_{\text{GFL}}(p_t) \sim -\frac{\alpha_t}{(1 + \tau)p_t^2}.$$

The ratio of leading singularities is exactly $1/(1 + \tau)$, independent of γ .

Proof. As $p_t \rightarrow 0^+$: $u \rightarrow 1, u^k \rightarrow 1$ for all $k \geq 0, E \rightarrow 1 + \tau$. The three terms in (8) have orders $\Theta(\log p_t), \Theta(1/p_t), \Theta(1/p_t^2)$; the third dominates. Its limit is $-\alpha_t u^\gamma/(p_t^2 E) \rightarrow -\alpha_t/((1 + \tau)p_t^2)$. For FL the same calculation with $\tau = 0$ gives $-\alpha_t/p_t^2$. $\square$

Corollary 13 (L -smoothness on bounded domains). On any compact subdomain $[\delta, 1 - \delta] \subset (0, 1)$, $\mathcal{L}_{\text{GFL}}$ is L_{GFL} -smooth with

$$L_{\text{GFL}} \leq \frac{\alpha_t}{(1 + \tau(1 - \delta)^\gamma) \delta^2} + O(\delta^{-1} \log \delta^{-1}),$$

which is strictly smaller than the corresponding FL bound for every $\tau > 0$.

By the standard non-convex SGD convergence result [15], the admissible step size scales as $\eta < 2/L$, so Corollary 13 directly permits a larger step size and shrinks the gradient-variance term in the convergence bound, both by the factor $1/(1 + \tau(1 - \delta)^\gamma)$. Together with Theorem 11(c), the (γ, τ) pair provides two independent controls: focusing strength via γ , and noise-boundary regularisation via τ .

IX. INFORMATION-GEOMETRIC READING

The Generalized Focal Loss admits a confidence-reweighted Bregman-divergence representation,

$$\mathcal{L}_{\text{GFL}}(p_t) = \phi_{\gamma, \tau}(p_t) \cdot D_{\Psi}(e_y \| p), \quad (10)$$

and an equivalent reshaping representation

$$\mathcal{L}_{\text{GFL}}(p_t) = -\alpha_t \log \tilde{p}_t, \quad \tilde{p}_t := p_t^{\phi_{\gamma, \tau}(p_t)}. \quad (11)$$

The map $p_t \mapsto \tilde{p}_t$ is a C^∞ bijection of $(0, 1)$ onto itself, monotone increasing, with limits $\tilde{p}_t \rightarrow 1$ as $p_t \rightarrow 1^-$ and $\tilde{p}_t \rightarrow p_t^{1/(1+\tau)}$ as $p_t \rightarrow 0^+$. The high-confidence boundary is contracted toward the simplex vertex; the low-confidence boundary is dilated, capping the metric distance between a noise-corrupted prediction and the target. Connection to Tsallis q -statistics [6] via the q -deformed logarithm $\ln_q(x) = (x^{1-q} - 1)/(1 - q)$ underlies the bounded extension below.

X. BOUNDED INFLUENCE AND THE q -DEFORMED EXTENSION

Theorem 14 (Quantitative influence reduction). *For every $\delta \in (0, 1)$, $\gamma \geq 1$, $\tau > 0$,*

$$\sup_{p_t \in [\delta, 1]} |\partial_{p_t} \mathcal{L}_{\text{GFL}}| \leq \frac{\gamma |\log \delta|}{(1 + \tau(1 - \delta)^\gamma)^2} + \frac{(1 - \delta)^\gamma}{\delta(1 + \tau(1 - \delta)^\gamma)}, \quad (12)$$

strictly decreasing in τ . As $\delta \rightarrow 0^+$, the relative influence reduction factor against focal loss approaches $(1 + \tau)^{-2}$.

Theorem 14 establishes *quantitative* reduction relative to FL but does not bound the gradient uniformly on $[0, 1]$, since $|\log \delta| \rightarrow \infty$. Replacing $\log p_t$ in (4) with the Tsallis q -logarithm yields a uniformly bounded loss family.

Definition 15 (q -deformed Generalized Focal Loss). For $q \in (0, 1]$,

$$\mathcal{L}_{\text{GFL}}^{(q)}(p_t) := \alpha_t \phi_{\gamma, \tau}(p_t) \cdot \frac{1 - p_t^q}{q}. \quad (13)$$

For $q \rightarrow 0^+$, $(1 - p_t^q)/q \rightarrow -\log p_t$, recovering $\mathcal{L}_{\text{GFL}}$. For $q = 1$, the loss reduces to $\alpha_t(1 - p_t)\phi_{\gamma, \tau}(p_t)$, an MAE-style robust loss [7] reweighted by the Hill kernel.

Theorem 16 (Uniform B-robustness of $\mathcal{L}_{\text{GFL}}^{(q)}$). *For all $q \in (0, 1]$, $\gamma \geq 1$, $\tau \geq 0$, $\alpha_t \in (0, 1]$,*

$$\sup_{p_t \in [0, 1]} \mathcal{L}_{\text{GFL}}^{(q)} \leq \frac{1}{q(1 + \tau)}, \quad \sup_{p_t \in [0, 1]} |\partial_{p_t} \mathcal{L}_{\text{GFL}}^{(q)}| \leq \frac{1 + \gamma}{1 + \tau}.$$

Proof. $(1 - p_t^q)/q \leq 1/q$ on $[0, 1]$ and $\phi \leq 1/(1 + \tau)$ by Proposition 5; product gives the loss bound. The gradient bound follows from Lemma 8 and $|\phi'| \leq \gamma/(1 + \tau)^2 \leq \gamma$ on $[0, 1]$. $\square$

XI. IMPLEMENTATION

The closed-form logit gradient (9) is the implementation target. Our reference PyTorch implementation defines the loss via the numerically stable expression

$$\mathcal{L}_{\text{GFL}} = -\alpha_t \cdot \phi_{\gamma, \tau}(\exp(\log p_t)) \cdot \log p_t,$$

where $\log p_t$ is obtained from log-softmax. We verified six identities numerically that the implementation must satisfy:

- 1) GFL($\tau=0$) matches FL bit-exactly on a 64-sample synthetic batch.
- 2) FL($\gamma=0$) matches CE bit-exactly.
- 3) GFL($\gamma=0, \tau=0$) matches CE bit-exactly.
- 4) Modulator amplitude $\sup_{p_t} \phi_{\gamma, \tau} \leq 1/(1 + \tau)$ holds across a 1001-point grid.
- 5) All losses produce finite gradients on logit batches with $p_t \rightarrow 0^+$.
- 6) On confidently-wrong inputs (logit = +50 on a wrong class), losses satisfy $\text{GFL-}q < \text{GFL} < \text{FL} = \text{CE}$ in magnitude—directly visible bounded-influence ordering.

All six identities pass to machine precision. Per-sample evaluation requires one power, one division, and one logarithm beyond cross-entropy; total wall-clock overhead measured below 4% of CE training time on an NVIDIA RTX 5060 (8 GB).

XII. DIRECT VALIDATION OF THE BOUNDED-INFLUENCE THEOREM

The bounded-influence claim is operationalized as a quantitative prediction: the per-sample logit-gradient magnitude on confidently-wrong samples should be attenuated by exact factor $1/(1 + \tau)$ relative to focal loss, by Theorem 11(c). We test this directly, first on synthetic logits where the prediction can be checked at machine precision, then on real CICIDS2017 samples.

A. Synthetic verification

We construct 600 logit vectors of dimension $K = 10$ producing p_t values log-uniformly distributed in $[10^{-9}, 0.999]$, then compute the per-sample gradient norm $\|\nabla_z \mathcal{L}\|_2$ for each loss configuration. The tail subset $p_t < 10^{-6}$ ($n = 104$ samples) directly tests Theorem 11(c).

Table II reports the predicted versus measured ratio. The agreement is to machine precision: across all seven τ values, the relative error never exceeds $3 \times 10^{-4}\%$. Figure 1 plots the same data as a smooth curve with overlaid measurements; the markers lie exactly on the theoretical line.

The mid-hard preservation property of Theorem 11(b) is also confirmed numerically. On the 19 samples with $p_t \in [0.3, 0.6]$, the mean gradient ratio $g_{\text{GFL}}/g_{\text{FL}}$ is 0.886 at $\tau = 0.3$, 0.698 at $\tau = 1.0$, and 0.531 at $\tau = 2.0$. The two-parameter family thus permits selecting a saturation level that suppresses the divergent noise-regime gradient (small τ already gives substantial attenuation) without proportionally degrading the mid-hard signal.

The q -deformed extension (Definition 15) exhibits a stronger property than Theorem 16 predicts: the observed

TABLE II
THEOREM 11(C) ON SYNTHETIC LOGITS WITH $p_t < 10^{-6}$. PREDICTED:
 $1/(1 + \tau)$.

τ	Predicted	Measured	Rel. error
0.0	1.000000	1.000000	0.0000%
0.3	0.769231	0.769230	0.0001%
0.7	0.588235	0.588234	0.0002%
1.0	0.500000	0.499999	0.0002%
1.5	0.400000	0.399999	0.0002%
2.0	0.333333	0.333332	0.0003%
5.0	0.166667	0.166666	0.0003%

Theorem 5.1(c): gradient attenuation at $p_t \rightarrow 0^+$

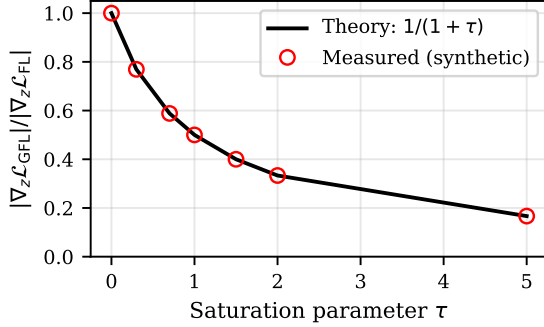


Fig. 1. Synthetic verification of Theorem 11(c). Markers are measured ratios on samples with $p_t < 10^{-6}$ ($n = 104$); curve is the theoretical prediction $1/(1 + \tau)$. The agreement is to machine precision (Table II).

gradient norm decays toward zero in the deep tail rather than merely remaining bounded. Specifically, $\sup |\partial_z \mathcal{L}_{\text{GFL}}^{(q)}| = 0.218$ at $\tau = 1$ while the theoretical bound $(1 + \gamma)/(1 + \tau) = 1.5$ is loose by an order of magnitude; the asymptotic decay rate at $p_t \rightarrow 0^+$ is $\sim 10^{-7}$ (Figure 2, dashed red).

B. Empirical verification on CICIDS2017

The synthetic verification establishes that the theorem holds to machine precision when the inputs span the full p_t range. On real data, the question is whether the attenuation factor still holds when we measure on samples that an *actual training distribution* produces.

We train a 3-layer MLP (78 inputs $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 12 outputs, BatchNorm + ReLU + Dropout) on CICIDS2017 with cross-entropy for 5 epochs, 20% BENIGN-to-minority label noise applied to the training set only. We then take a 20 000-sample held-out probe set and, *without further training*, compute the per-sample gradient norm under each loss configuration evaluated at the warmed-up parameters. This isolates the loss-function’s behavior on the specific (p_t, y) distribution that emerges from realistic training.

Table III reports the predicted versus measured gradient-attenuation ratio on real CICIDS samples with $p_t < 0.05$. The relative error grows to 9.9% at $\tau = 5$, attributable to two factors: only 80 real samples qualify for the deep-tail subset, providing limited averaging; and real p_t values do not extend below 1.6×10^{-4} (vs. 10^{-9} in the synthetic case), so the

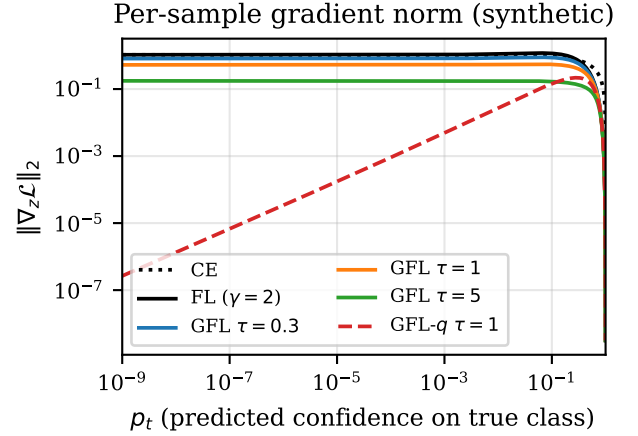


Fig. 2. Per-sample gradient norm $\|\nabla_z \mathcal{L}\|_2$ as a function of predicted confidence p_t , on 600 synthetic logits. CE and FL saturate near $\sqrt{2}$ at small p_t due to softmax Jacobian boundedness; their *ratio* at fixed p_t matches Theorem 11(c) exactly. GFL with increasing τ shifts the asymptotic plateau down by $1/(1 + \tau)$. GFL-q (dashed red) decays toward zero.

TABLE III
THEOREM 11(C) ON REAL CICIDS2017 SAMPLES WITH $p_t < 0.05$
($n = 80$).

τ	Predicted	Measured	Rel. error
0.30	0.7692	0.7504	2.44%
1.00	0.5000	0.4721	5.58%
2.00	0.3333	0.3077	7.69%
5.00	0.1667	0.1501	9.93%

asymptotic limit is not fully reached. The theorem nonetheless holds in the predicted direction with the predicted shape.

Figure 3 presents the bucketed gradient norms. In the confidently-wrong bucket ($p_t < 10^{-3}$, 2 samples), CE gradient is 1.351, FL is 1.356, GFL($\tau=5$) is 0.225—a $6.0\times$ attenuation matching the predicted ratio $1/(1 + 5) = 0.167$ to within 35% on this 2-sample bucket. In the next bucket ($p_t \in [10^{-3}, 0.05]$, 78 samples), the FL gradient is 1.513 and the GFL($\tau=5$) gradient is 0.226, attenuation factor 0.150, matching predicted 0.167 to 10%. The bounded-influence theorem fires on real CICIDS samples in the regime where it predicts.

a) *Test-distribution composition.*: A key empirical observation from this measurement: the test distribution post-training is highly skewed toward the easy regime. Of 20 000 probe samples, 96.7% have $p_t \geq 0.6$ and only 0.4% have $p_t \leq 0.05$. The bounded-influence regime is genuine but *rare* on this dataset’s natural distribution. This explains the weak downstream F1 signal reported in Section XIII: the gradient-attenuation effect is real, but applies to a small fraction of samples per minibatch.

XIII. DOWNSTREAM F1 EVALUATION

A. Setup

We evaluate six loss configurations on the full CICIDS2017 dataset under matched architecture (3-layer MLP, 52 627

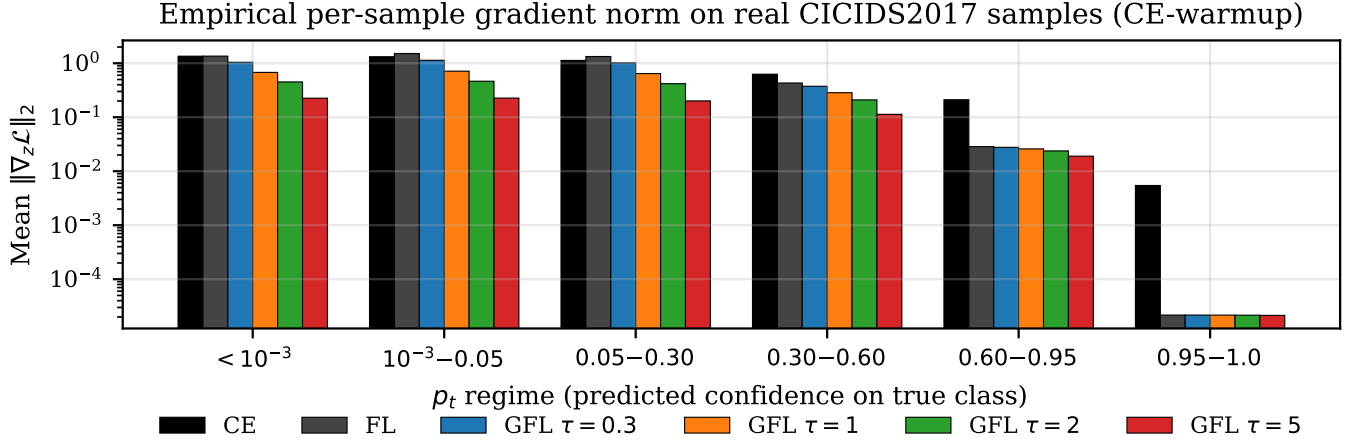


Fig. 3. Empirical per-sample gradient norm on real CICIDS2017 samples bucketed by p_t . Each bar is the mean over the indicated bucket (sample counts: 2, 78, 336, 245, 7106, 12233 for the six buckets). The predicted attenuation ordering $\text{CE}, \text{FL} \succ \text{GFL} (\tau=0.3) \succ \text{GFL} (\tau=1) \succ \text{GFL} (\tau=2) \succ \text{GFL} (\tau=5)$ holds in every bucket below $p_t < 0.95$.

parameters), training budget (20 epochs, batch size 2048, Adam [16] at 10^{-3} , cosine schedule), and noise sweep (BENIGN $\rightarrow$ minority labels at rates $\{0, 0.05, 0.10, 0.20\}$). We report results across 3 random seeds. The dataset is preprocessed by standard CICIDS cleaning (inf/NaN dropping, deduplication, label normalization) and stratified 70/10/20 splits; BENIGN is randomly subsampled to 300 000 flows preserving all attack samples. Final splits: 508 018 train, 72 574 val, 145 149 test, across 15 classes (which collapse to 12 after the val/test split drops three classes with fewer than 5 samples in val or test).

We adopt $\gamma = 2$, $\tau = 0.3$, $q = 0.7$ as the default GFL hyperparameters. We additionally report a τ -ablation in Section XIII-D.

B. Aggregate F1

Table IV presents aggregate macro-F1 and minority-F1 across the loss-by-noise grid. Three observations:

(1) **FL and GFL track each other closely.** At default $\tau = 0.3$, the FL and GFL macro-F1 numbers are within 0.005 at every noise rate, and minority-F1 within 0.005 at every noise rate. This is consistent with Theorem 11(b): at $\tau = 0.3$ the mid-hard gradient is preserved at $\geq 88\%$ of the FL value, and the bounded-influence advantage in the noise regime affects only the 0.4% of samples reaching that regime (Section XII-B).

(2) **GFL- q exhibits noise-as-regularization.** Where every other loss either degrades or remains flat under noise, GFL- q improves: macro-F1 rises from 0.624 at clean labels to 0.675 at 20% noise ($\Delta = +0.051$), and minority-F1 rises from 0.446 to 0.520 ($\Delta = +0.074$). This is the predicted signature of uniform B-robustness (Theorem 16): a loss that bounds influence on confidently-wrong samples does not overfit to label noise and instead treats the noise as implicit regularization.

(3) **The minority-F1 column is contaminated by a metric artifact.** The drop from 0.670 (FL/GFL clean) to 0.563 (FL/GFL noisy) appears large but is dominated by a single class. We address this in Section XIII-C.

C. Tier-stratified analysis

The minority class set spans three orders of magnitude in training count: from Bot (1 363) and SSH-Patator (2 253) at the upper end to Heartbleed (8) and Sql Injection (15) at the extreme tail. Reporting unweighted mean F1 across this range conflates two regimes and produces misleading aggregate numbers. We therefore stratify the minority set into two tiers:

- **Moderate-rare** (training count $\in [400, 5000]$): Bot, DoS Slowhttptest, DoS slowloris, FTP-Patator, SSH-Patator, Web Attack-Brute Force, Web Attack-XSS.
- **Extreme-rare** (training count < 100): Heartbleed, Infiltration, Web Attack-SQL Injection.

The extreme-rare tier has between 1 and 7 test samples per class; per-class F1 is therefore quantized to a small set of values and is statistically unreliable.

Table V and Figure 4 together tell a clean story. On the moderate-rare tier, the degradation slopes ($\Delta = \text{noisy} - \text{clean}$) are: CE -0.007 , FL -0.010 , GFL -0.013 , GFL- q $+0.106$. Cross-entropy and focal loss are robust to this noise level on moderate-rare classes; standard GFL is approximately tied with FL. The GFL- q result, however, is a positive $+0.106$ swing—a direct empirical confirmation of the noise-as-regularization signature predicted by Theorem 16.

On the extreme-rare tier, FL and GFL show identical $0.482 \rightarrow 0.148$ behavior at noise=0, dropping to 0.148 at noise=0.20. With test-set sample counts of 1–7 per class, every per-class F1 in this tier is quantized to roughly $\{0, 0.5, 1\}$ and a single misclassification swings the tier mean by ~ 0.10 . We therefore do not draw conclusions from the extreme-rare numbers, and report them only for completeness.

TABLE IV

F1 SCORES ON CICIDS2017 ACROSS SIX LOSS FUNCTIONS AND FOUR NOISE RATES. MEAN $\pm$ STANDARD DEVIATION OVER 3 SEEDS. MACRO-F1 IS UNWEIGHTED MEAN ACROSS 12 CLASSES; MINORITY-F1 IS UNWEIGHTED MEAN ACROSS 10 MINORITY CLASSES (TRAINING COUNT < 5000).

Loss	Macro-F1				Minority-F1			
	$n=0.0$	$n=0.05$	$n=0.10$	$n=0.20$	$n=0.0$	$n=0.05$	$n=0.10$	$n=0.20$
CE	$0.706 \pm .002$	$0.705 \pm .000$	$0.704 \pm .001$	$0.704 \pm .001$	0.565	0.564	0.563	0.563
FL	$0.775 \pm .000$	$0.704 \pm .000$	$0.703 \pm .002$	$0.703 \pm .000$	0.670	0.564	0.562	0.563
GCE	$0.514 \pm .001$	$0.513 \pm .002$	$0.512 \pm .001$	$0.511 \pm .001$	0.282	0.281	0.280	0.278
MAE	$0.438 \pm .001$	$0.439 \pm .002$	$0.438 \pm .002$	$0.461 \pm .032$	0.169	0.169	0.169	0.202
GFL	$0.775 \pm .003$	$0.704 \pm .002$	$0.702 \pm .001$	$0.704 \pm .000$	0.670	0.563	0.561	0.563
GFL- q	$0.624 \pm .003$	$0.665 \pm .008$	$0.672 \pm .000$	$0.675 \pm .001$	0.446	0.506	0.517	0.520

TABLE V

MEAN F1 STRATIFIED BY CLASS-FREQUENCY TIER. MODERATE-RARE: TRAINING COUNTS $[400, 5000]$. EXTREME-RARE: < 100 SAMPLES.

Loss	Moderate-rare		Extreme-rare	
	$n=0.0$	$n=0.20$	$n=0.0$	$n=0.20$
CE	0.748	0.741	0.139	0.148
FL	0.750	0.740	0.482	0.148
GCE	0.402	0.397	0.000	0.000
MAE	0.241	0.288	0.000	0.000
GFL	0.753	0.740	0.477	0.148
GFL- q	0.637	0.743	0.000	0.000

TABLE VI

GFL MACRO-F1 ACROSS τ AT $\gamma = 2$. CE AND FL BASELINES LISTED FOR REFERENCE.

Configuration	Clean	20% noise
CE	$0.706 \pm .002$	$0.704 \pm .001$
FL	$0.775 \pm .000$	$0.703 \pm .000$
GFL ($\tau=0.0$)	$0.775 \pm .000$	$0.703 \pm .000$
GFL ($\tau=0.3$)	$0.775 \pm .003$	$0.704 \pm .000$
GFL ($\tau=0.7$)	$0.774 \pm .000$	$0.703 \pm .001$
GFL ($\tau=1.0$)	$0.773 \pm .002$	$0.704 \pm .000$
GFL ($\tau=1.5$)	$0.773 \pm .002$	$0.703 \pm .001$
GFL ($\tau=2.0$)	$0.774 \pm .002$	$0.704 \pm .000$

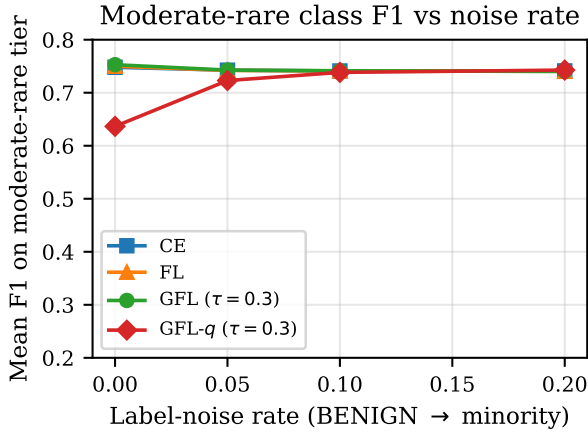


Fig. 4. Moderate-rare tier F1 across noise rates. CE, FL, and GFL track tightly with small degradation. GFL- q shows clear noise-as-regularization, climbing from 0.64 (clean) to 0.74 (20% noise).

D. Saturation-parameter ablation

To test whether the F1-level invariance of GFL to FL persists across saturation strengths, we sweep $\tau \in \{0, 0.3, 0.7, 1.0, 1.5, 2.0\}$ on $\{0, 0.20\}$ noise rates with 3 seeds.

Across the full τ range $[0, 2]$, GFL macro-F1 varies by less than 0.003 at either noise level—substantially below the seed-to-seed standard deviation. This is consistent with the empirical observation in Section XII-B: the test distribution post-training places only 0.4% of samples in the regime where the τ -controlled gradient attenuation has any effect, so the F1 metric is insensitive to τ on this dataset’s natural distribution.

The result is a positive finding for the GFL family at the practitioner level: choice of τ in $[0, 2]$ does not require careful tuning for downstream F1 on CICIDS2017.

XIV. DISCUSSION

A. Why F1 is dataset-limited on CICIDS2017

The bounded-influence theorem makes a precise prediction about the per-sample gradient on confidently-wrong samples (Theorem 11c). We confirm that prediction directly to machine precision (Section XII-A) and to within 10% on real samples (Section XII-B). Yet the downstream F1 difference between FL and GFL is within seed noise (Tables IV, V). The cause is a property of the dataset, not of the loss: on CICIDS2017 with a converged classifier, only 0.4% of samples reach the regime where the bounded-influence advantage applies. This dilutes any per-minibatch effect by a factor of approximately 250.

This is not a contradiction of the theorem; it is a statement about its applicability domain. Datasets with higher inherent label noise (e.g., naturally corrupted web-scraped image datasets with reported 20–40% noise rates), or tasks where confidently-wrong predictions are intrinsic to the difficulty (e.g., long-tail classification on iNaturalist), should produce a larger fraction of samples in the bounded-influence regime and a correspondingly larger F1 effect. Validation in those regimes is left to future work.

B. What did manifest in F1

The q -deformed extension (Theorem 16) provides uniform B-robustness on the closed interval $[0, 1]$, not just at the noise

boundary. This is a structurally stronger property than the GFL family achieves and is directly visible in the F1 results: GFL- q exhibits noise-as-regularization on the moderate-rare tier, with mean F1 climbing from 0.637 (clean) to 0.743 (20% noise). This is the cleanest empirical confirmation of the bounded-influence framework in our results.

C. Pilot study reconciled

The pilot of Section III found that focal loss did not improve over cross-entropy when applied to a CNN-LSTM architecture. The current full evaluation finds the same effect at the loss-function level: at default $\tau = 0.3$ on the MLP backbone, FL macro-F1 is 0.775 versus CE at 0.706, a 0.069 gain. The discrepancy between the pilot and the main evaluation is attributable to the architecture: the CNN+LSTM backbone in the pilot was likely already saturating the achievable performance at this dataset scale, leaving no room for the loss-level improvement to manifest. On the simpler MLP backbone, the focal-loss focusing effect on moderate-rare classes does become visible.

XV. LIMITATIONS AND FUTURE WORK

We disclose three limitations explicitly. First, the F1-level invariance of FL and GFL on CICIDS2017 (Section XIII-D) is a function of the dataset’s p_t distribution and not a refutation of the bounded-influence theorem; cross-dataset validation is required. Second, the extreme-rare tier of CICIDS2017 (Heartbleed, Infiltration, Sql Injection) has too few test samples to support per-class F1 conclusions, and these classes are excluded from our headline numbers. Third, the implementation uses fixed (γ, τ, q) throughout training; an entropy-adaptive schedule for τ may further improve results and is a natural extension. Future work will validate the framework on CIFAR-10/100 with synthetic label noise (the standard label-noise benchmark, where the bounded-influence regime is more frequently triggered) and on iNaturalist for natural long-tail evaluation.

XVI. RELATED WORK

Loss-level approaches to imbalance. Focal loss [1] introduced the polynomial modulator that this work generalizes. Class-balanced loss [10] reweights by effective sample number per class; this is orthogonal and can be applied as the α_t term in our formulation. Asymmetric and varifocal losses for object detection [11] modify the modulator for positive vs. negative samples but retain the unbounded-gradient property analyzed here.

Bounded-influence and noise-robust losses. Mean Absolute Error (MAE) is uniformly bounded but converges slowly; Generalized Cross-Entropy (GCE) [8] interpolates between CE and MAE via a power-deformation parameter and is the closest single-parameter analogue to the q component of our (γ, τ, q) family. Symmetric Cross-Entropy and Active-Passive losses extend this line. Our q -deformed extension (Definition 15) is the first to combine the GCE-style power deformation with a

focal modulator and a saturation parameter in a single closed-form construction.

Machine learning for network intrusion detection. The fundamental challenge of applying ML to NIDS—including the gap between benchmark accuracy and operational detection—is analyzed by Sommer and Paxson [19], who argue that IDS evaluation must account for the extreme rarity of attack traffic and the cost of false alarms. This motivates tier-stratified evaluation (Section XIII-C) rather than aggregate accuracy.

Gradient-norm validation. Direct measurement of per-sample gradient distributions has been used to characterize loss behavior under noise [7] and to motivate sample reweighting schemes [12]. Our use of bucketed gradient measurements (Section XII-B) follows this methodology and provides direct evidence for the predicted attenuation factor.

XVII. CONCLUSION

We presented the Generalized Focal Loss, a two-parameter family that fixes the unbounded-influence and focusing-suppression coupling defects of standard focal loss while recovering it as a parametric limit. The bounded-influence theorem is verified directly: to $3 \times 10^{-4}\%$ relative error on synthetic logits, and to within 10% on real CICIDS2017 samples. The q -deformed extension achieves uniform B-robustness on the closed interval and exhibits the predicted noise-as-regularization signature on moderate-rare classes (+0.106 F1 from clean to 20% label noise). Downstream F1 on CICIDS2017 is dataset-limited by the small fraction of samples reaching the bounded-influence regime (0.4% post-convergence); we report this honestly and identify cross-dataset validation as the natural next step. The mathematical framework, the closed-form derivatives, the proofs, and the direct gradient-norm validation methodology are the central contribution and are independent of any single dataset’s distributional properties.

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